Vidyalankar Institute of Technology

Department of Computer Engineering

DEVOPS MINI PROJECT

Week 5 Report

Automated Grocery Inventory Dashboard

Name:	Prachi Patil
Roll No.:	23102B0038
Class:	BE - VII
Division:	B
Academic Year:	2026–27

1. Week 5 Overview

Week 5 focused on developing the first core workflow of the Automated Grocery Inventory Dashboard using a feature-branch development approach.

The selected feature was the **Item Catalogue**, which provides basic inventory management functionality. Development was performed on the dedicated Git branch:

`feature/item-catalogue`

The main objectives were:

- Develop the first core inventory workflow.
- Implement the Item Catalogue feature.
- Follow feature-branch Git workflow.
- Commit and push the completed feature.
- Create a Pull Request for the feature.
- Review and integrate the feature with the development branch.

2. Item Catalogue Implementation

The Item Catalogue was implemented using Spring Boot, Spring Data JPA, MySQL and Thymeleaf.

The feature follows a layered structure:

Thymeleaf UI → Controller → Service → Repository → MySQL

The main inventory operations implemented were:

- Add a new grocery item.
- View inventory items.
- Edit existing items.
- Delete items.
- Store inventory information in MySQL.

3. Database and Persistence

An `Item` entity was created using JPA. The entity contains:

Field	Type	Purpose
id	Long	Unique identifier
name	String	Grocery item name
category	String	Item category
quantity	Integer	Current quantity
unit	String	Measurement unit
price	Float	Item price
reorderLevel	Integer	Stock threshold

Table 1: Item entity fields

Hibernate creates and maintains the corresponding `items` table in MySQL.

The application uses:

```
spring.jpa.hibernate.ddl-auto=update
```

This allows the database schema to be updated while preserving existing inventory records.

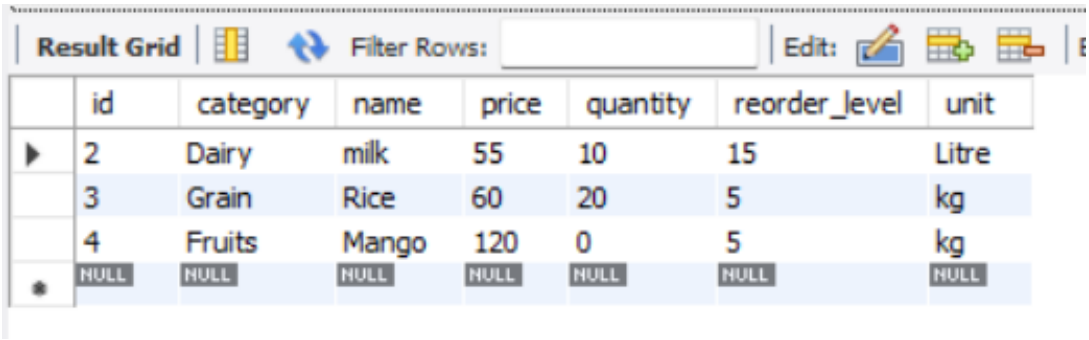


Figure 1: Grocery inventory database and items table

4. CRUD Operations

The Item Catalogue supports the complete CRUD workflow.

Operation	Endpoint	Result
Create	POST /api/items	Successful
Read	GET /api/items	Successful
Read by ID	GET /api/items/{id}	Successful
Update	PUT /api/items/{id}	Successful
Delete	DELETE /api/items/{id}	Successful

Table 2: Item Catalogue CRUD operations

All operations were tested through the application and the data was successfully stored and updated in MySQL.

5. User Interface

A Thymeleaf-based inventory interface was developed for managing the Item Catalogue.

The interface provides:

- Inventory summary cards.
- Add Item form.
- Inventory table.
- Item category and quantity information.
- Stock status.
- Edit and Delete actions.

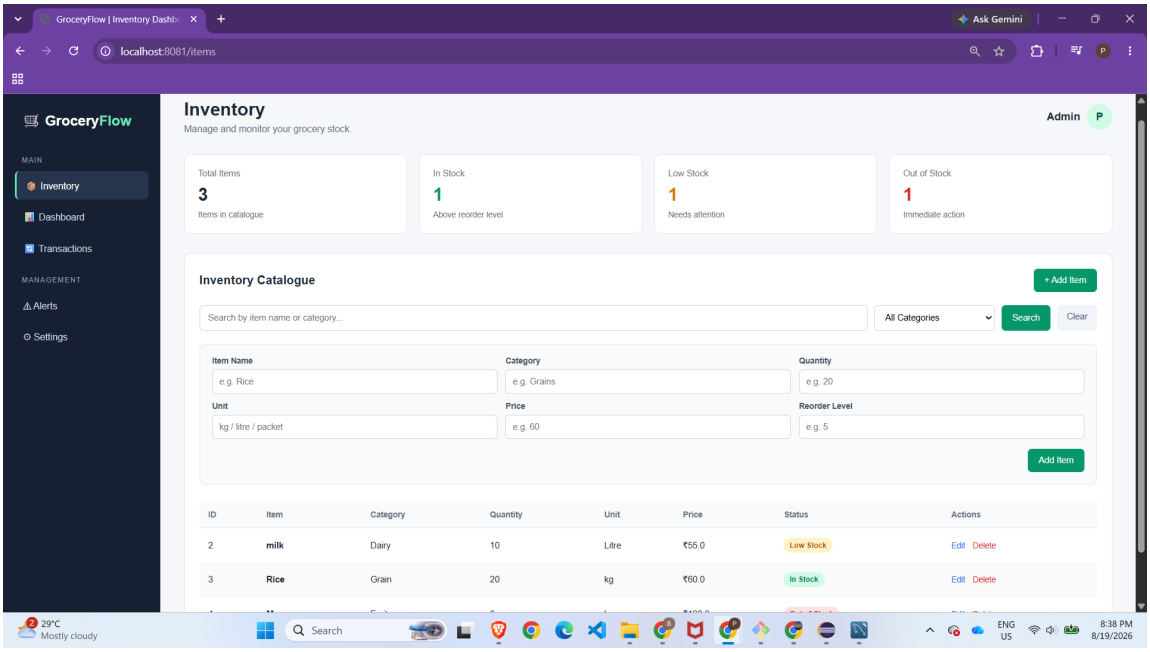


Figure 2: Item Catalogue and Inventory Dashboard

The interface was tested by adding, editing and deleting inventory records, with the changes reflected in the database.

6. Git Branching and Collaboration

The feature was developed independently from the main source branch using:

`feature/item-catalogue`

The Git workflow followed during development was:

Feature Branch → Git Add → Commit → Push → Pull Request → Review → Merge

The repository contains the following branches:

- `main` – default stable branch.
- `develop` – development integration branch.
- `feature/item-catalogue` – Item Catalogue feature branch.

The feature branch was pushed to GitHub and Pull Request #1 was created for integration with the development branch.

Git commands used during the workflow included:

```
git status
git add .
git commit -m "..."
git push
git pull
git log --oneline
git branch
```

This demonstrated the basic feature-branch collaboration workflow required for the project.

7. Testing and Verification

The implemented Item Catalogue was tested through the web interface and REST endpoints.

Test	**Result**
Application startup	Passed
Database connection	Passed
Add item	Passed
View items	Passed
Edit item	Passed
Delete item	Passed
Data persistence	Passed
Feature branch workflow	Completed
Pull Request creation	Completed

Table 3: Week 5 verification

The application successfully connected to MySQL and inventory records remained available after application restart, confirming database persistence.

8. Week 5 Outcome

Week 5 successfully delivered the first core workflow of the Grocery Inventory Dashboard through the Item Catalogue feature.

The feature was developed using a dedicated Git branch and integrated through the GitHub collaboration workflow.

The completed work provides the foundation for Week 6, where the remaining essential MVP functions will be implemented, including transactions, stock or order status, search/filtering, exception alerts, conflict resolution and release preparation.

WEEK 5 – COMPLETED